Tarea 2

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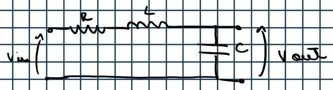
1) Transferencia Pasa-bajos de orden $n = 6$.



$$\begin{cases} T(s) = \dfrac{Z_2}{Z_1 + Z_2} & (1) \\ Z_2 = 1/sC & (2) \\ Z_1 = Z_s = R + sL & (3) \end{cases}$$

$$T(s) = \frac{1/sC}{R + sL + 1/sC} = \frac{1}{s^2 . LC + s.RC + 1}$$

Normalizando la expresión:

$$T(s) = \frac{1/L.C}{s^2 + s.R/L + 1/L.C}$$

y expresando en función de Q y w_0:

$$T(s) = \frac{w_0^2}{s^2 + s.\frac{w_0}{Q} + w_0^2}$$

SOS n $\longrightarrow$ Se utiliza solo para los pasa-bajos.

al ser de orden $n = 6 \longrightarrow$ tres circuitos RLC en serie. Suma de tres transferencias.

$$T(s) = \frac{K_1 w_{01}^2}{s^2 + s.\frac{w_{01}}{Q_1} + w_{01}^2} + \frac{K_2 w_{02}^2}{s^2 + s\frac{w_{02}}{Q_2} + w_{02}^2} + \frac{w_{03}^2 . K_3}{s^2 + s\frac{w_{03}}{Q_3} + w_{03}^2}$$

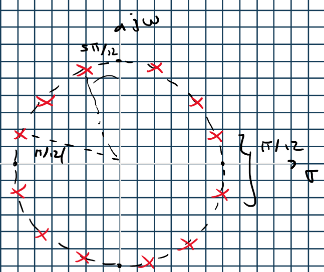
K permite ajustar el nivel de la banda de paso. $K \in \mathbb{R}$

Es una cte. que trabaja con la ganancia.

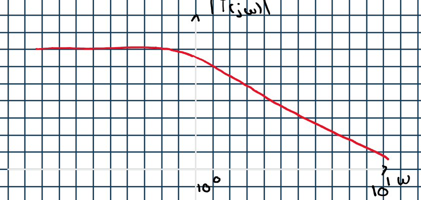
2) Diagrama de polos y ceros:

como $n = 6 \longrightarrow$ 6 pares de polos c.c en el diagrama.

$\frac{\pi}{2n} = \frac{\pi}{12} \longrightarrow$ distancia entre polos

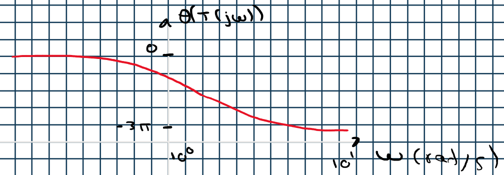


Análisis de Rta. de módulo

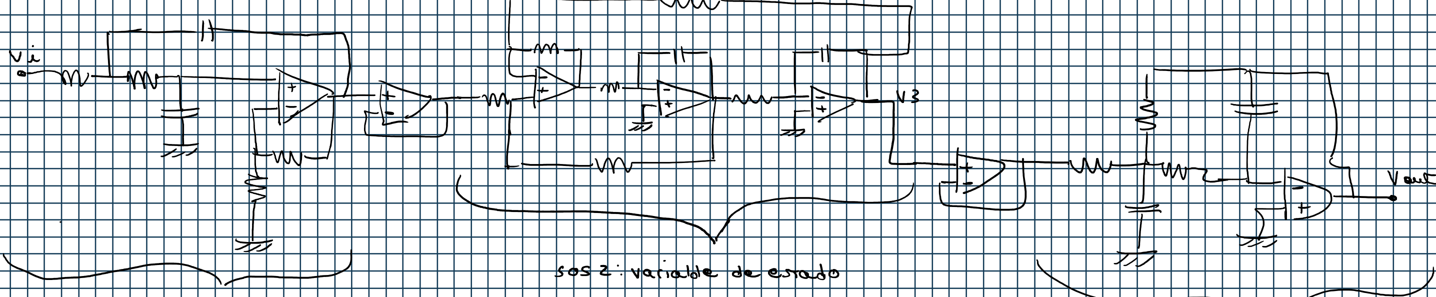


Por ser Butterworth:

$\epsilon = 1 \longrightarrow$ en w_0, $|T|^2$ es $1/2$



3)



SOS 1: Sallen-Key

SOS 2: variable de estado

conecto solo el pasa-bajo. Es la salida V3 del circuito.

SOS 3: multiple feedback

criterio de diseño:

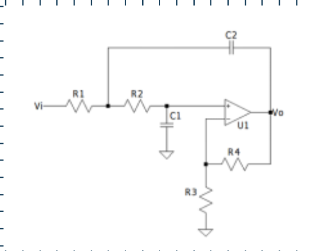
$Q_1 < Q_2 < Q_3$.

Recuperando la expresión calculada en el punto anterior.

$$T(s) = \frac{K_1\,\omega_{01}^2}{s^2 + s\,\frac{\omega_{01}}{Q_1} + \omega_{01}^2} + \frac{K_2\,\omega_{02}^2}{s^2 + s\,\frac{\omega_{02}}{Q_2} + \omega_{02}^2} + \frac{\omega_{03}^2\,K_3}{s^2 + s\,\frac{\omega_{03}^2}{Q_3} + \omega_{03}^2} \;//$$

Recuperando las expresiones de normalización con doble escalado:

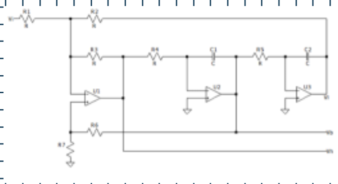
$$\begin{cases} R = \Omega_z \\ L = L'' \cdot \frac{\Omega_z}{\Omega_\omega} \\ C = C'' / \Omega_z \cdot \Omega_\omega \end{cases} \longrightarrow \text{partiendo de } \Omega_z = 1 \text{ y } \Omega_\omega = \omega_0 .$$



Transferencia del Sallen-Key:

$$H(s) = \frac{K / R_1.R_2.C_1.C_2}{s^2 + s.\frac{(R_1.C_1 + R_2.C_2 + R_1.C_1.(1-K))}{R_1 R_2.C_1.C_2} + \frac{1}{R_1 R_2.C_1.C_2}}$$

$$\omega_{01} = \frac{1}{\sqrt{R_1.R_2.C_1.C_2}}$$



Transferencia del filtro variable de estado: planteando la transferencia tomando solo la salida V_L (Low):

$$T_2(s) = \frac{-R_3 / R_1.R_4.C_1.R_5.C_2}{s^2 + s.\left(\frac{1}{R_4.C_1}\cdot\left(\frac{R_7}{R_6+R_7}\right)\cdot\left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2}\right)\right) + \frac{R_3}{R_2.R_4.C_1.R_5.C_2}}$$

Por simplicidad, nuevamente expresamos que:

$R_1 = R_2 = R_3 = R_4 = R_5 = R$
$C_1 = C_2 = C$

$$\longrightarrow \frac{\cancel{R}}{R^{\cancel{3}\,2}.C^2} = \omega_{02}^2$$

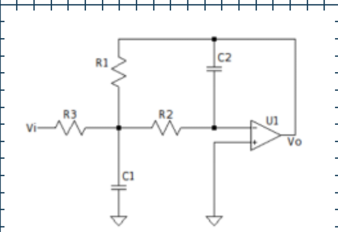
$$\frac{1}{R^2.C^2} = \omega_{02}^2$$

$$\omega_{02} = \frac{1}{R.C}$$

$$\frac{\omega_{02}}{Q} = \frac{1}{R.C}\cdot\left(\frac{R_7}{R_6+R_7}\right)\cdot(3)$$

$$\begin{cases} \frac{\omega_{02}}{Q} = \frac{3.R_7}{R_6+R_7} & (1) \\ \omega_{02} = \frac{1}{R.C} & (2) \end{cases}$$

$$Q_2 = \frac{1}{3}\cdot\left(1 + \frac{R_6}{R_7}\right)$$



Transferencia del filtro multiple feed-back:

$$T_3(s) = \frac{K.\,1/R_1.R_2.C_1.C_2}{s^2 + s\left(\frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)\right) + \frac{1}{R_1.R_2.C_1.C_2}}$$

$$K = -\frac{R_1}{R_3}$$

- Filtro Sallen-Key:

$R = \Omega_z = 1$ y $\Omega_\omega = \omega_{01} = 1$

$$f_c = \frac{1}{2\pi R.C} = 0{,}159\,Hz$$

$$\begin{cases} Q = \frac{1}{2.\cos \pi/12} = 0{,}5176 \approx 0{,}518 \\ R_1 = R_2 = 1 = R_3 \longrightarrow \frac{3-K}{R.C} = \frac{\omega_0}{Q} \\ C_1 = C_2 = C = 1 \\ \omega_0 = 1 \qquad -K = \frac{1}{Q} - 3 \\ K = 1 + R_4 \end{cases}$$

$\hookrightarrow R_4 = 0{,}07$ $\qquad K = 3 - 1/Q \longrightarrow K = 1{,}07 //$

- Filtro variable de estado:

$$\begin{cases} Q_2 = \frac{1}{3}\cdot\left(1 + \frac{R_6}{R_7}\right) \longrightarrow Q_2 = \frac{1}{2.\cos\left(\frac{3\pi}{12}\right)} = 0{,}7071 \\ R_1 \text{ a } R_6 = R = 1 \\ \omega_0 = 1 \\ C = 1 \end{cases}$$

$$Q_2 = \frac{\sqrt{2}}{2} = \frac{1}{3}\cdot\left(1 + \frac{1}{R_7}\right)$$

$$\frac{3.\sqrt{2}}{2} - 1 = 1 / R_7 = 1{,}121$$

$$\frac{1}{R_7} = 1{,}121 \longrightarrow R_7 = 0{,}892 //$$

- Filtro multiple Feedback:

$$Q = \frac{1}{2.\cos(5\pi/12)} = 1{,}9318$$

$$\begin{cases} \frac{\omega_0}{Q} = \frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \\ \text{la ganancia es } K = -\frac{R_1}{R_3} \end{cases}$$

criterios de normalización:

$\omega_0 = 1$ rad/s

$R_3 = R_1 = R_2 = R \longrightarrow 1\Omega.$

$$\omega_0 = \frac{1}{R.\sqrt{C_1.C_2}} \longrightarrow C_1 \cdot C_2 = 1$$

$$\frac{1}{Q} = C_2.(\;3\;)$$

$$\frac{1}{Q} = 3.C_2 \longrightarrow C_2 = 0{,}172 //$$

$$C_1 = 5{,}814 //$$

$\frac{3.\sqrt{2}}{2} - 1 = 1 / R7 = 1,121$

$\frac{1}{R7} = 1,121 \longrightarrow R7 = 0,892$ //

$\frac{1}{Q} = 3.C2 \longrightarrow C2 = 0,172$ //

$C1 = 5,814$ //

4) $\begin{cases} R = R2 = 1 \\ \Omega w_0 = w_0 = 1 \text{ rad/s} \end{cases}$

en caso de querer ganancia 10 dB.

Ganancia actual:

$K1.K2.K3 = 1.07 \cdot 1,121 \cdot 1 = |1,19947| \approx |1,2|$

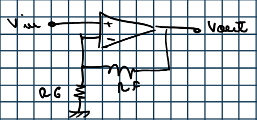
(producto de las tres etapas)

$10\,dB \longrightarrow 20.\log_{10}(x) = 10\,dB$

$10^{10/20} = 3,1622 \longrightarrow$ valor al que quiero llegar

$\frac{3,1622}{1,12} = 2,823$

Uso un Buffer no Inversor para subir la ganancia y lograr la banda de paso solicitada



$A = 1 + \frac{RF}{RG} \longrightarrow \frac{RF}{RG} = 1,823$

$RG = R2 = 1$

$RF = 1,823$ //

$A = 2,823 \quad K_{TOT} = 3,3876 \longrightarrow 10,6\,dB$ //